

Unit 27: The Fundamental Theorem of Calculus — Full Solutions

Warm-up

1. $\int_0^2 x^2 dx$

Antiderivative: $F(x) = \frac{x^3}{3}$.

$$F(2) - F(0) = \frac{8}{3} - 0 = \frac{8}{3}$$

Answer: $\frac{8}{3}$.

2. $\int_1^3 (2x + 1) dx$

Antiderivative: $F(x) = x^2 + x$.

$$F(3) - F(1) = (9 + 3) - (1 + 1) = 12 - 2 = 10$$

Answer: 10.

3. $\int_0^4 x dx$

Antiderivative: $F(x) = \frac{x^2}{2}$.

$$F(4) - F(0) = 8 - 0 = 8$$

Answer: 8.

4. $\int_{-1}^2 3x^2 dx$

Antiderivative: $F(x) = x^3$.

$$F(2) - F(-1) = 8 - (-1) = 9$$

Answer: 9.

5. $\int_0^1 (x^3 + x) dx$

Antiderivative: $F(x) = \frac{x^4}{4} + \frac{x^2}{2}$.

$$F(1) - F(0) = \left(\frac{1}{4} + \frac{1}{2}\right) - 0 = \frac{3}{4}$$

Answer: $\frac{3}{4}$.

6. $\int_1^2 4 dx$

Antiderivative: $F(x) = 4x$.

$$F(2) - F(1) = 8 - 4 = 4$$

Answer: 4.

7. $\int_0^\pi \sin x dx$

Antiderivative: $F(x) = -\cos x$.

$$F(\pi) - F(0) = (-\cos \pi) - (-\cos 0) = -(-1) - (-1) = 1 - (-1) = 2$$

Answer: 2.

Standard

8. $\int_0^{\pi/2} \cos x dx$

Antiderivative: $F(x) = \sin x$.

$$F\left(\frac{\pi}{2}\right) - F(0) = 1 - 0 = 1$$

Answer: 1.

9. $\int_1^4 \sqrt{x} dx$

Rewrite $\sqrt{x} = x^{1/2}$. Antiderivative: $F(x) = \frac{2}{3}x^{3/2}$.

$$F(4) - F(1) = \frac{2}{3}(8) - \frac{2}{3}(1) = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}$$

Answer: $\frac{14}{3}$.

10. $\int_1^2 \frac{1}{x^2} dx$

Rewrite $\frac{1}{x^2} = x^{-2}$. Antiderivative: $F(x) = -x^{-1} = -\frac{1}{x}$.

$$F(2) - F(1) = -\frac{1}{2} - (-1) = -\frac{1}{2} + 1 = \frac{1}{2}$$

Answer: $\frac{1}{2}$.

11. $\int_0^2 (x^2 - 2x + 3) dx$

Antiderivative: $F(x) = \frac{x^3}{3} - x^2 + 3x$.

$$F(2) - F(0) = \left(\frac{8}{3} - 4 + 6\right) - 0 = \frac{8}{3} + 2 = \frac{8}{3} + \frac{6}{3} = \frac{14}{3}$$

Answer: $\frac{14}{3}$.

12. $\frac{d}{dx} \left[\int_2^x (t^2 + 1) dt \right]$

By FTC Part 1, this is simply the integrand evaluated at x :

$$\frac{d}{dx} \left[\int_2^x (t^2 + 1) dt \right] = x^2 + 1$$

13. $\frac{d}{dx} \left[\int_0^x \sin t dt \right]$

By FTC Part 1:

$$\frac{d}{dx} \left[\int_0^x \sin t dt \right] = \sin x$$

14. $\int_{-2}^2 x^3 dx$

Antiderivative: $F(x) = \frac{x^4}{4}$.

$$F(2) - F(-2) = \frac{16}{4} - \frac{16}{4} = 4 - 4 = 0$$

Answer: 0 — this confirms exactly what we'd predict from Unit 26's odd-function symmetry short-cut, since x^3 is an odd function and $[-2, 2]$ is symmetric about 0.

Challenge

15. $\frac{d}{dx} \left[\int_1^{x^2} (t^3 + 1) dt \right]$

Using the chain rule version of FTC Part 1, with $g(x) = x^2$ (so $g'(x) = 2x$):

$$\frac{d}{dx} \left[\int_1^{x^2} (t^3 + 1) dt \right] = [(x^2)^3 + 1] \cdot 2x = (x^6 + 1)(2x) = 2x^7 + 2x$$

Answer: $2x^7 + 2x$.

$$16. \frac{d}{dx} \left[\int_0^{\sin x} \sqrt{1+t^2} dt \right]$$

Using the chain rule version, with $g(x) = \sin x$ (so $g'(x) = \cos x$):

$$\frac{d}{dx} \left[\int_0^{\sin x} \sqrt{1+t^2} dt \right] = \sqrt{1+\sin^2 x} \cdot \cos x$$

Answer: $\sqrt{1+\sin^2 x} \cdot \cos x$.

$$17. \int_1^2 \frac{x^4 + x^2}{x^2} dx$$

Simplify first by dividing each term by x^2 :

$$\frac{x^4}{x^2} + \frac{x^2}{x^2} = x^2 + 1$$

Now integrate: antiderivative $F(x) = \frac{x^3}{3} + x$.

$$F(2) - F(1) = \left(\frac{8}{3} + 2 \right) - \left(\frac{1}{3} + 1 \right) = \left(\frac{8}{3} + \frac{6}{3} \right) - \left(\frac{1}{3} + \frac{3}{3} \right) = \frac{14}{3} - \frac{4}{3} = \frac{10}{3}$$

Answer: $\frac{10}{3}$.

$$18. \int_0^3 |x-1| dx$$

Split at $x = 1$, where the absolute value changes behavior: for $x < 1$, $|x-1| = 1-x$; for $x \geq 1$, $|x-1| = x-1$.

$$\int_0^3 |x-1| dx = \int_0^1 (1-x) dx + \int_1^3 (x-1) dx$$

First piece: antiderivative $x - \frac{x^2}{2}$.

$$\left[1 - \frac{1}{2} \right] - [0 - 0] = 0.5$$

Second piece: antiderivative $\frac{x^2}{2} - x$.

$$\left[\frac{9}{2}-3\right]-\left[\frac{1}{2}-1\right]=[1.5]-[-0.5]=2$$

Total:

$$0.5+2=2.5$$

Answer: 2.5.

$$19. F(x) = \int_0^x (t^2 - 4) dt$$

By FTC Part 1:

$$F'(x) = x^2 - 4$$

Set $F'(x) = 0$: $x^2 - 4 = 0 \Rightarrow x = \pm 2$.

Use the First Derivative Test on $F'(x) = (x-2)(x+2)$:

Test $x < -2$ (e.g., -3): $(-5)(-1) = 5 > 0 \rightarrow F$ increasing.

Test $-2 < x < 2$ (e.g., 0): $(-2)(2) = -4 < 0 \rightarrow F$ decreasing.

Test $x > 2$ (e.g., 3): $(1)(5) = 5 > 0 \rightarrow F$ increasing.

Answer: F has a local maximum at $x = -2$ (increasing then decreasing) and a local minimum at $x = 2$ (decreasing then increasing).

Applied

$$20. v(t) = 3t^2 - 6t \text{ on } [0, 3].$$

$$\text{Antiderivative: } F(t) = t^3 - 3t^2.$$

$$F(3) - F(0) = (27 - 27) - 0 = 0$$

Answer: net displacement = 0 — the particle ends up back where it started, even though it moved during the interval.

$$21. r(t) = 6t + 2 \text{ on } [0, 5].$$

$$\text{Antiderivative: } F(t) = 3t^2 + 2t.$$

$$F(5) - F(0) = (75 + 10) - 0 = 85$$

Answer: total water collected = 85 gallons.

$$22. MC(x) = 2x + 10, \text{ from } x = 5 \text{ to } x = 10.$$

$$\text{Antiderivative: } F(x) = x^2 + 10x.$$

$$F(10) - F(5) = (100 + 100) - (25 + 50) = 200 - 75 = 125$$

Answer: total additional cost = \$125.

23. $D(t) = \int_0^t v(s) ds$, where $v(s) = 4s + 1$.

By FTC Part 1:

$$D'(t) = v(t) = 4t + 1$$

Answer: $D'(t) = 4t + 1$, which represents the object's instantaneous velocity at time t — the rate at which total distance is accumulating at that exact moment.